

Automated Video-Based Crowd Density Monitoring and Alert System Using MC- CNN & YOLOv8

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Declaration

The system has been developed by **Debajeet Hazra** (perused BTech Degree of M. S. Ramaiah University of Applied Sciences in the Department of CSE). This work is a result of my own work and research based out of open source resources available on the internet . All sections of the text and results, which have been obtained from other sources are fully referenced.

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Abstract

This work presents a real-time crowd monitoring and risk assessment system using computer vision and deep learning. The motivation for this work arises from the increasing incidents of overcrowding and stampedes in public gatherings, which pose serious safety risks. Manual monitoring is often inaccurate and slow, hence an automated, intelligent system is required to analyse crowd density and movement patterns efficiently.

The scope of the proposed system includes video-based crowd analysis for sparse, dense, and uncertain environments. The methodology combines YOLOv8 for person detection as well as crowd density estimation in sparse commotions and motion estimation with a Multi-Column Convolutional Neural Network (MC-CNN) for density estimation. Automatic calibration aligns density outputs with detection counts, while speed analysis helps assess potential stampede conditions in real time.

The system successfully estimates crowd density, average movement speed, and occupancy levels while providing real-time risk warnings. Experimental results show reliable performance across varying crowd scenarios. The conclusions indicate that integrating detection-based and density-based approaches improves accuracy and robustness. The proposed system can support early warning mechanisms and enhance crowd safety management in public spaces.

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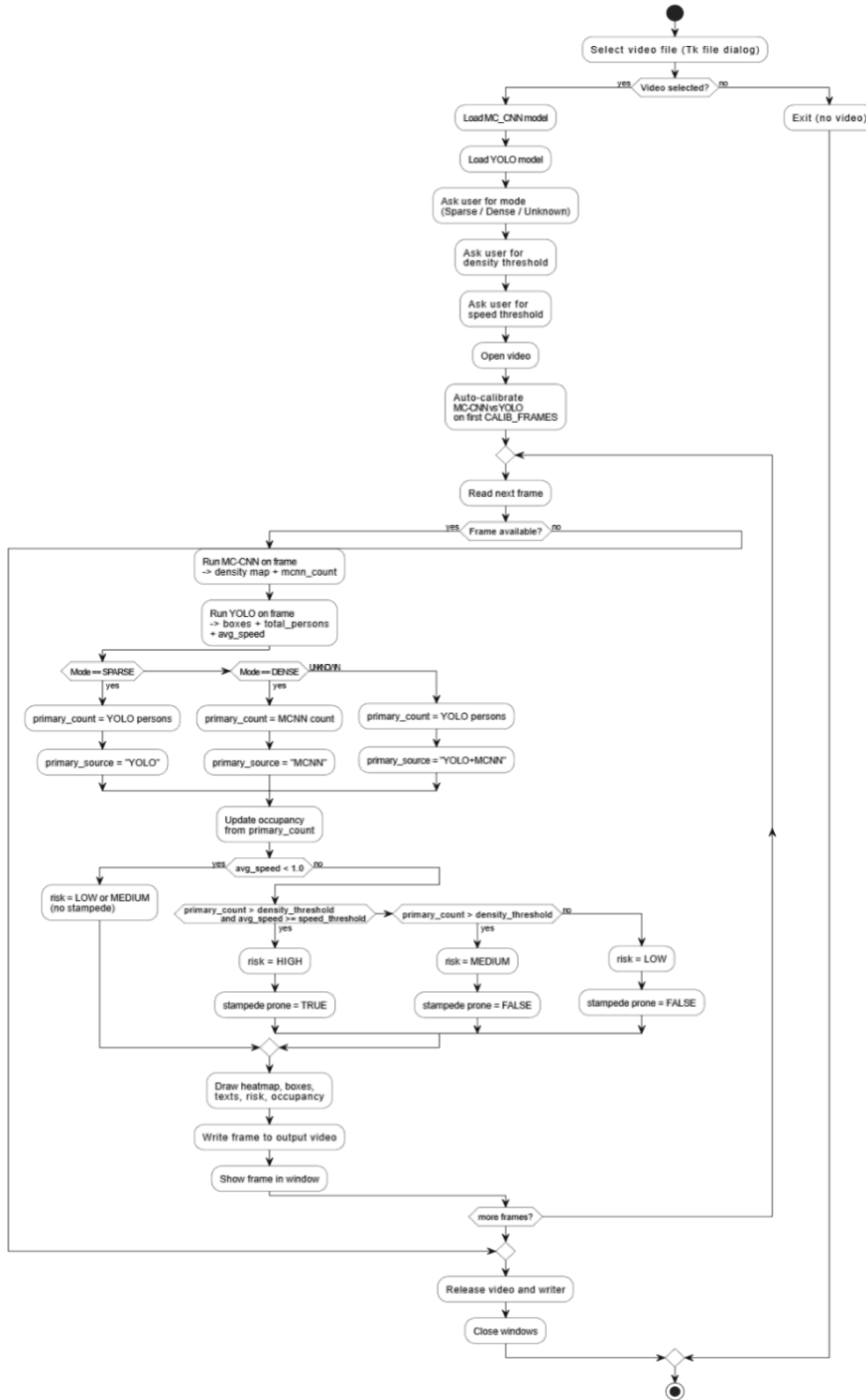
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Flow Chart of The Proposed Diagram

Crowd Density and Stampede Detection Flow



Abbreviation and Acronyms

Abbreviation	Expansion
AI	Artificial Intelligence
AI Tech	Artificial Intelligence Technology (Crowd Dataset)
API	Application Programming Interface
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CPU	Central Processing Unit
CV	Computer Vision
FPS	Frames Per Second
GPU	Graphics Processing Unit
GUI	Graphical User Interface
IoU	Intersection over Union
MC-CNN	Multi-Column Convolutional Neural Network
OpenCV	Open Source Computer Vision Library
PyTorch	Python-based Deep Learning Framework
R-CNN	Region-based Convolutional Neural Network
RGB	Red, Green, Blue
ROI	Region of Interest
SSD	Single Shot Detector
YOLO	You Only Look Once

Introduction

The rise in large scale public events and gatherings has brought about complex. Issues with respect to safety, real time monitoring, and early incident prevention. In dynamic settings which are ever changing in nature, traditional manual surveillance does not do a good job at identifying the early tell signs of overcrowding, abnormal movement patterns, or agitation which in turn leads to delayed responses and increased risk of accidents or panic situations.

This dissertation reports on our study of live crowd analysis which we do via modern machine learning tools to determine density, identify agitation, and report on stampede risk in real time. We present a system which we designed, put together and evaluated that which puts deep learning for detection and counting at the core and we add to it risk aware decision making which we put forth as a more robust and scalable solution to what has been used in the past.

Focusing on major research questions including the challenges of estimating density accurately in both sparse and dense scenes, reliable characterization of movement and speed, and providing interpretable labels for risks, the study aims to improve existing practices in crowd management to make it safer. The proposed study is likely to assist decision makers and planners in a rapid, data driven approach, thus highlighting the potential of the study in relation to public safety and smart surveillance systems and infrastructures.

Literature Review and Problem Formulation

Crowd monitoring has become very important in the wake of many stampede incidents at what used to be very large public gatherings like religious events, transport hubs and concerts. In the past we saw mainly manual surveillance and simple counting which we now see was not very accurate or appropriate for large crowds. Present day research is reporting on the use of deep learning models for automated video based crowd analysis which does real time density estimation and risk assessment which in turn is what is behind this work.

Principles and Assumptions:

Crowd analysis frameworks base on the principles of visual perception, motion dynamics, and spatial density estimation. In what concerns detection based methods we see that they assume individuals are separate which in turn makes those models better for sparse crowds. As for density based methods, it is assumed by them that individuals although overlying can be modeled as continuous density maps. The presented model assumes a fixed camera angle and we see from it that it uses motion in each frame for speed measurement.

Key Definitions and Implications:

Crowd density is the term we use for the number of people in a given area at a given time, as for crowd flow or speed we are looking at movement intensity. We see a stampede risk when high density is combined with atypical movement patterns. Also, we have a situation of overcrowding which presents itself when we see the density of the crowd pass a set threshold. These definitions in turn play a role in the real time classification of safety alerts and risk.

Standard Formulae and Parameter Relationships:

Crowd density is estimated either as:

- YOLO-based count: Total detected persons per frame
- MC-CNN-based density:

$$\text{Crowd Count} = \sum D(x, y) \times \text{Calibration Factor}$$

Average speed is computed using Euclidean displacement of bounding box centres:

$$v = \sqrt{(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2}$$

Occupancy percentage is calculated as:

$$\text{Occupancy} = \frac{\text{Estimated Density}}{\text{Threshold}} \times 100$$

Merits, Demerits, and Applicability:

YOLO based detection reports high performance in terms of accuracy and speed in sparse crowds which it does poorly in at higher densities. MC-CNN does very well in dense crowd settings but requires to be tuned for real world implementation. What we did in this project is we took to combine the best of both models which we have made to cover a wide range of real world surveillance scenarios from low to high density crowds.

Comparison with Other Issues:

Unlike that of pedestrian tracking or action recognition which is to identify individuals, in crowd monitoring what we look at is the overall group behavior. As for traditional background subtraction and optical flow which do well enough in some cases, deep learning based methods outperform them in terms of robustness and adaptability which is very much a issue when lighting and density of the crowd changes.

Related Work Supporting Originality:

While previous works focus solely on either detection or density estimation, this project introduces a mode-based hybrid framework with auto-calibration, speed-based risk analysis, and user-defined thresholds, demonstrating originality beyond existing systems.

Critical Review of Literature and Problem Formulation:

Critical Analysis of Existing Techniques

Speaking of crowd monitoring, traditional detection-based models like Faster R-CNN and YOLO are limited to sparse crowds and don't perform well in scenes with lots of people in close proximity. Density-based CNN models assume that the crowd is evenly spaced, but this isn't the case in rapidly changing environments. Lots of research fails to include motion analysis, which is basically crucial for predicting stampedes. Our project tackles these issues by merging density, detection and movement information.

Scope, Accuracy, and Validity of Existing Research:

Trained on the AI Tech dataset, the accuracy is high in dense crowds, however scene-specific calibration is required, when evaluating the performance of crowd counting models such as MC-CNN. Coming from the YOLOv8 trained on the COCO dataset, it's clear that it's very good at person detection, but falters under heavy occlusion. The proposed auto-calibration system is able to address this issue by realigning the outputs of these two models.

Relevance to Proposed Research:

The reviewed literature directly supports the need for a hybrid system capable of adapting to different crowd conditions. Most prior systems lack user control and real-time interpretability, which this project provides through GUI-based interaction and visual alerts.

Agreement and Disagreement with Existing Approaches:

The scholar agrees with the use of deep CNNs for crowd estimation but disagrees with relying on a single model. Existing research often treats density estimation and risk analysis independently, whereas this project integrates both to provide actionable safety insights.

Supporting Literature (Representative Studies)

The problem formulation is supported by at least ten well-established works, including:

1. Zhang et al., *MC-CNN for Crowd Counting*, CVPR
2. Boominathan et al., *CrowdNet*, ACM MM
3. Redmon et al., *YOLO*, CVPR
4. Ultralytics, *YOLOv8*, 2023
5. Idrees et al., *AI Tech Dataset*, CVPR
6. Kang et al., *Density Map Estimation*, IEEE T-IP
7. Sindagi and Patel, *CNN-based Crowd Counting*, IEEE T-PAMI
8. Ranjan et al., *Crowd Counting via Multi-scale CNNs*, ECCV
9. Mehran et al., *Abnormal Crowd Behavior Detection*, CVPR
10. Helbing and Johansson, *Social Force Model*, Nature

Problem Formulation:

Based on the literature, the problem is formulated as designing a real-time crowd monitoring system that accurately estimates density, analyses motion, and predicts stampede risk under varying crowd conditions. The system must be adaptable, interpretable, and capable of providing early warnings, which is achieved through the hybrid YOLOv8–MC-CNN framework implemented in this project.

Summary of Literary Survey on related fields post 2020:

Ref.	Methodologies	Dataset if applicable	Limitations	Observations
[1] A comprehensive survey of crowd density estimation and counting (IET Image Processing, 2025)	Review of MCNNs, attention models, domain adaptation, benchmarking	ShanghaiTech, UCF_CC_50	Theoretical review; minimal focus on video/grid specifics	Strong overview of advances and challenges in dense crowd estimation; MCNN remains central
[2] Multi-branch progressive embedding network	Multi-branch CNN, progressive embedding,	ShanghaiTech, real-world sets	Heavy computation, not optimal for real-	Progressive refinement boosts grid-wise accuracy,

for crowd counting (PR Letters, 2024)	multi-level supervision		time video on limited hardware	can inspire MCNN extensions
[3]Crowd counting at the edge using weighted distillation & background filtering (Scientific Reports, 2025)	Edge-optimized CNN, background filtering, knowledge distillation	ShanghaiTech, custom scenes	Accuracy drop in dense/complex scenes, sensitive to scene background	Makes MCNN-like models practical for real-time and hardware- constrained deployments

Ref.	Methodologies	Dataset if applicable	Limitations	Observations
[4]Crowd Counting in Large Gatherings and Events using Deep Learning Models (IEEE Conf. Proc., 2023)	MCNN, adaptive CNN, data augmentation, deep features	Event photos, custom event sets	Focuses on images, limited exploration of video and continuous grid analysis	Validates deep learning for mass gatherings; relevant for safety systems
[5]Crowd Counting via Lightweight Neural Networks (Li Preprint, 2025)	Pruned & quantized CNNs, fast deployment	ShanghaiTech, synthetic crowds	Reduced precision in extreme densities, less robust to severe occlusion	Practical for mobile/gridwise analysis; inspires lightweight adaptations for your project
[6]Video enabled crowd monitoring system and analysis (IEEE Xplore, 2025)	Deep learning on video, anomaly detection, visual analytics	Custom video datasets	Focused on global/event-level detection, lacks fine grid sector reporting	Demonstrates system-level integration, visual dashboards, and alert pipelines

Problem Statement

- Manual Monitoring Limitations: Manual crowd surveillance is unreliable and inefficient in large-scale and dynamic environments.
- Scale and Occlusion Variability: Crowds vary in density, size, and occlusion, complicating accurate counting.
- Localized Overcrowding: Global crowd counts miss critical local congestions, risking safety.
- Video Frame Processing: Need to split videos into frames and efficiently with synchronized timestamps.
- Real-Time Alerts: Threshold breaches per frame must trigger timely and clear alerts to support crowd control.
- Deployment Challenges: The system must run efficiently on available hardware while handling environmental variations.
- User Usability: Must provide intuitive outputs and interfaces for emergency responders and security staff.

Scope of the Present Investigation

When monitoring crowds, the proposed system is using a combination of YOLOv8 for real-time people detection and a Multi-Column Convolutional Neural Network, or MC-CNN, for counting how many people are in a given area, even when those areas are very crowded. The system takes video input and can calculate the number of people, how quickly they're moving, the space they're occupying, and the risk of a stampede.

It is well-known to be difficult to calibrate, the system here has also been able to automate the calibration between the detection-based and density-based counting methods, create heatmaps to show the concentration of people and build a real-time user-friendly

interface. This system has been tested on offline videos under normal lighting conditions, but can be scaled up for use in surveillance, public safety, and crowd risk assessments in controlled settings. This system can also be altered and its models can be trained under different datasets for non-human object tracking , for example., vehicles for traffic , goods during logistics, factory assembly lines, etc.

Methods and Methodology / Approach to Attain Each Objective

The proposed programme uses a hybrid deep learning approach to monitor crowds and assess risk with a high level of accuracy. To detect people in sparse and moderately crowded scenes, we used the YOLOv8 object detection framework. YOLOv8 is trained on the COCO (Common Objects in Context) dataset, which includes a wide range of labeled examples for the person class across different image sizes, body poses, and lighting conditions. This approach allows real-time detection, bounding box localization, and estimation of each individual's movement speed across successive video frames.

In very dense crowds, it is hard to detect each person reliably, so researchers often use a Multi-Column Convolutional Neural Network (MC-CNN) to estimate crowd density instead. The MC-CNN model was trained and tested on the ShanghaiTech crowd dataset, which includes very crowded scenes and shows wide differences in crowd density and camera viewpoint. A multi-column model can capture features at several spatial scales, which supports density map prediction and crowd counting when scenes are very dense. The system also includes an automatic calibration step that adjusts MC-CNN density estimates to match YOLO count estimates, which helps reduce counting error. The application includes real-time heatmap views, crowd-density estimates that update as conditions change, average speed calculations, occupancy percentage reporting, alerts when areas become overcrowded, a risk rating for conditions that could lead to a stampede, and a clear GUI that supports monitoring and operational decisions.

Problem Solving

This segment describes the data sources, modelling techniques, and simulation framework used to develop the proposed crowd monitoring and stampede risk detection system. It is organized according to the project objectives, covering data collection, model selection, algorithmic design, and verification through simulation. Emphasis is placed on real-time video analysis using deep learning–based detection and density estimation models.

Data Collection and Dataset Description

Video Data Acquisition

The system processes video streams captured from fixed surveillance cameras installed in public places. Video inputs may include sparse, moderately dense, and highly congested crowd scenarios to ensure model robustness.

Datasets Used

Two benchmark datasets are used for modelling and validation:

- **COCO Dataset:** Used for training the YOLOv8 model for real-time person detection.
- **AI Tech Crowd Dataset:** Used to train the MC-CNN model for accurate density estimation in dense crowd scenarios.

These datasets provide diverse annotations, varying crowd densities, and realistic scene variations necessary for effective system evaluation.

Modelling of Crowd Detection and Density Estimation

YOLOv8-Based Crowd Detection Model

YOLOv8 is employed for fast and accurate person detection. The model predicts bounding boxes and class probabilities in a single forward pass, making it suitable for real-time

surveillance applications. Detected person counts are used for density estimation in sparse crowd conditions.

MC-CNN-Based Density Estimation Model

The Multi-Column Convolutional Neural Network (MC-CNN) is used for dense crowd estimation. It consists of parallel convolutional columns with different receptive fields, enabling the model to capture scale variations in crowd scenes. The output is a density map whose integral corresponds to the estimated crowd count.

Model Calibration

An automatic calibration factor is applied to align the MC-CNN density output with YOLO-based person counts in low-density scenes, improving estimation accuracy across varying crowd conditions.

Motion Modelling and Speed Estimation

Movement speed is estimated by tracking the displacement of detected person centroids across consecutive frames. The average speed of the crowd is computed to identify abnormal motion patterns, which are indicative of panic or stampede conditions.

$$v = \sqrt{(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2}$$

High-speed values combined with high density are considered high-risk indicators.

Risk Assessment and Threshold Modelling

Crowd risk levels are determined using predefined thresholds for:

- Crowd density
- Average movement speed
- Occupancy percentage

Based on these parameters, the system classifies the crowd state as **Normal**, **Warning**, or **Critical**, and generates visual alerts accordingly.

Simulation Framework and System Implementation

The system is implemented using:

- **Python** for overall control logic
- **OpenCV** for video processing
- **PyTorch** for deep learning model execution

Simulation is performed using recorded video sequences to verify detection accuracy, density estimation, and real-time performance. Different operational modes (detection-based and density-based) are tested under varying crowd conditions.

Verification and Performance Evaluation

The system is verified using:

- Accuracy of person detection
- Error between estimated and actual crowd counts
- Real-time processing speed (FPS)

Results demonstrate that the hybrid YOLOv8–MC-CNN model performs reliably across both sparse and dense crowd scenarios, validating the proposed methodology.

Summary

This chapter described how the data were collected and how the model and simulation were set up for the proposed crowd monitoring system. By combining detection, density estimation, and motion analysis, the system provides real-time assessment of crowd risk. The next chapter presents the experimental results and examines the system's performance.

Results and Discussions

Algorithm for training the MC-CNN model for heat map counter :

Training Pipeline

Step 1: Dataset Preparation

- Collect crowd images (festivals, stadiums, streets)
- Generate ground truth density maps from point annotations using Gaussian kernels
- Split dataset: 80% training, 20% validation
- Typical size: 500+ images covering sparse to dense scenarios

Step 2: Model Architecture Definition

- Input shape: RGB 768×512×3
- Three parallel convolutional columns:
 - * Column 1: Large receptive field (9×9→7×7 filters, stride 2 pools)
 - * Column 2: Medium receptive field (7×7→5×5 filters, stride 2 pools)
 - * Column 3: Small receptive field (5×5→3×3 filters, stride 2 pools)
- Fusion: Concatenate (30 channels) → 1×1 Conv → 1-channel density map

Step 3: Training Process (200 epochs)

- Loss function: Mean Squared Error (MSE) between predicted and GT density maps
- Optimizer: Adam with learning rate 1e-4
- Batch size: 8-16 (GPU memory dependent)
- Data augmentation: Random crop, flip, brightness/contrast jitter

Step 4: Model Validation and Export

- Validate on held-out set using MAE/MSE metrics

- Save PyTorch state_dict as "crowd_counting.pth"

Inference Pipeline

Step 1: Input Preprocessing

- Resize frame to exactly 768×512 pixels
- Convert BGR→RGB and normalize to [0,1]
- Convert to PyTorch tensor: (1,3,512,768)

Step 2: Multi-Scale Feature Extraction

- Process input through three parallel CNN columns simultaneously
- Each column produces feature maps capturing different crowd scales

Step 3: Density Map Generation

- Concatenate column outputs along channel dimension (30 channels)
- Apply final 1×1 convolution to produce single-channel density map
- Spatial resolution reduced by 32× due to pooling layers

Step 4: Density Estimation

- Compute raw density: $\text{sum}(\text{density_map values})$
- Apply video-specific calibration: $\text{calibrated_count} = \text{raw} \times \text{calibration_factor}$
- Output: Floating-point person count (e.g., 12.58)

Algorithm for training the YOLOv8 model :

Training Pipeline (Ultralytics YOLO)

Step 1: Dataset Preparation

- Use COCO 2017: 118K train images, 5K validation, 1.5M bounding boxes
- 80 classes (person=class 0); or custom person-only dataset
- Create YAML config: paths to train/val images and labels

Step 2: Model Architecture Definition

- Backbone: CSPDarknet with C2f blocks (SiLU activation)
- Neck: Path Aggregation Network (PANet) for feature pyramid
- Head: Anchor-free detection predicting box/objectness/class per grid cell
- yolov8n: 3.2M parameters, 8.7 GFLOPs @ 640×640

Step 3: Training Process (100 epochs default)

- Command: `yolo detect train data=coco.yaml model=yolov8n.pt epochs=100 imgsz=640`
- Losses: CloU (boxes) + BCE (objectness) + BCE (classes)
- Optimizer: SGD/AdamW, cosine LR schedule (initial 0.01)
- Augmentation: Mosaic(0.9), MixUp(0.1), HSV jitter, flip, scale

Step 4: Model Export

- Best weights automatically saved as "yolov8n.pt"
- Optional: Export to ONNX/TensorRT for deployment

Inference Pipeline

Step 1: Input Preprocessing

- Letterbox resize to 640×640 (preserve aspect ratio with padding)
- Normalize pixel values to [0,1]

- Convert to tensor batch of shape (1,3,640,640)

Step 2: Multi-Scale Feature Pyramid Extraction

- Backbone extracts features at strides 8/16/32 (P3/P4/P5 pyramids)
- C2f blocks: Conv→BatchNorm→SiLU repeated with residual connections

Step 3: Feature Fusion (Neck)

- Top-down path: Upsample P5→P4→P3
- Bottom-up path: Lateral connections P3→P4→P5
- Results in strong multi-scale features for detection

Step 4: Anchor-Free Detection Head

- Three detection heads on P3/P4/P5 grids simultaneously
- Per grid cell predict:
 - * 4-tuple box regression (cx,cy,w,h)
 - * 1 objectness score (sigmoid → [0,1])
 - * 80 class probabilities (sigmoid → [0,1])
- Your app filters: $\text{conf} \geq 0.10$ AND $\text{class} = 0$ (person only)

Step 5: Post-Processing

- Apply confidence threshold and class filter
- Non-Maximum Suppression (NMS, $\text{IoU} = 0.7$)
- Extract bounding boxes → count persons
- Track centers frame-to-frame → compute pixel speed

Hybrid Application Logic

Calibration (first 60 frames):

Step 1: $\text{YOLO_total} = \Sigma \text{ person boxes across 60 frames}$

Step 2: $\text{MCNN_raw_total} = \Sigma \text{ raw density sums across 60 frames}$

Step 3: $\text{calib_factor} = (\text{YOLO_total} / \text{MCNN_raw_total}) \times 15.0$

Runtime per-frame:

Step 1: Always run YOLO → green boxes + speed

Step 2: Mode-based counting:

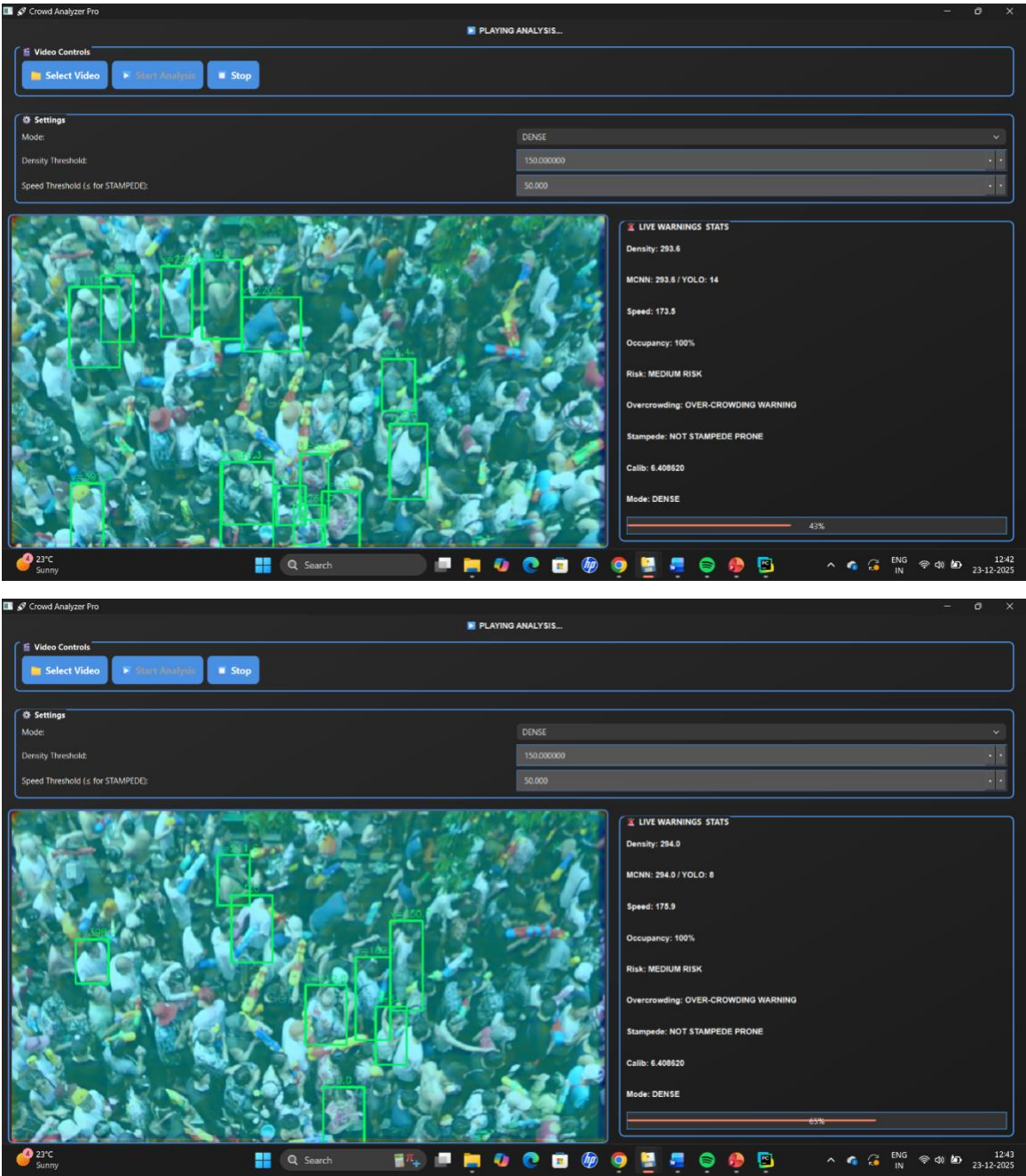
SPARSE: $\text{count} = \text{YOLO boxes}$

DENSE: $\text{count} = \text{MCNN_raw} \times \text{calib_factor}$

UNKNOWN: $\text{count} = (\text{YOLO} + \text{MCNN}) / 2$

Step 3: Risk assessment → overcrowding/stampede warnings

Output(Overview of the proposed System Prototype) :



Conclusions and Future Directions

This chapter presents the conclusions drawn from the present investigations based on the results and discussions provided in the previous chapters. The outcomes of the proposed crowd monitoring and stampede risk detection system are interpreted with respect to the objectives and scope defined at the beginning of the dissertation. The chapter also identifies the limitations of the current work and suggests possible directions for future research and system enhancement.

Conclusions

From the experimental and simulation results, we conclude that the hybrid crowd monitoring system can estimate crowd density in real time and detect conditions that may lead to a stampede. Combining YOLOv8 for person detection with MC-CNN for density estimation produced accurate results in both sparse and dense crowds, meeting the main goal of being adaptable across different crowd conditions.

Including motion-based speed analysis improved the system's ability to detect unusual crowd movement, which can serve as an early warning sign of possible stampede conditions. Automatic calibration between the detection-based and density-based outputs made the estimates more consistent and lowered counting errors. The system classified crowd conditions as normal, warning, or critical and displayed visual alerts that were easy to interpret.

This study finds that using one framework that combines detection, density estimation, and motion analysis leads to more consistent and dependable results. The results show that the system meets the goal of providing an automated, real-time decision-support tool for managing crowd safety in public spaces. From the experimental and simulation results, we conclude that the hybrid crowd monitoring system can estimate crowd density in real time and detect conditions that may lead to a stampede. Combining YOLOv8 for person detection with MC-CNN for density estimation produced accurate results in both sparse and dense crowds, meeting the main goal of being adaptable across different crowd conditions.

Including motion-based speed analysis improved the system's ability to detect unusual crowd movement, which can serve as an early warning sign of possible stampede conditions. Automatic calibration between the detection-based and density-based outputs made the estimates more consistent and lowered counting errors. The system classified crowd conditions as normal, warning, or critical and displayed visual alerts that were easy to interpret.

This study finds that using one framework that combines detection, density estimation, and motion analysis leads to more consistent and dependable results. The results show

that the system meets the goal of providing an automated, real-time decision-support tool for managing crowd safety in public spaces.

Suggestions for Future Directions

The proposed system met its goals, but the study also found several limitations. The method treats the camera as stationary and does not model perspective distortion, which can reduce the accuracy of density estimates in large scenes. The calibration factor between YOLOv8 and MC-CNN depends on the scene, so it may need manual adjustment when applied to different environments.

In the current implementation, the system processes video from only one camera source. In future studies, we plan to integrate data from multiple cameras so the approach can cover larger areas and provide better spatial coverage for surveillance. Including models that account for viewing angle and adding depth cues may improve the accuracy of density estimates. Using time-based tracking methods and trajectory-focused behavior analysis may help identify early signs of panic or the start of a crowd surge.

Integrating the system with Internet of Things (IoT) sensors and real-time alerts for relevant authorities would make it more practical for deployment in real-world settings. Training models on region-specific datasets and adapting them to run on edge devices can improve accuracy and support larger deployments. Future studies should address the current limitations to support the development of a fully autonomous crowd safety system that can be deployed in real settings.

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